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 Studierichting: Informatica  
 Oefeningenreeks 3G nr. 2

$$f(x) = (x - 2)^3 + 1$$

$$f(x) = x^3 - 6x^2 + 12x - 7$$

$$\begin{array}{c|ccc|c} 1 & 1 & -6 & 12 & -7 \\ & \downarrow & 1 & -5 & 7 \\ \hline & 1 & -5 & 7 & 0 \end{array}$$

$$f(x) = (x - 1)(x^2 - 5x + 7)$$

$$D = (-5)^2 - 4 \cdot 1 \cdot 7 = -3 \Rightarrow \text{geen nulpunten}$$

$$f'(x) = 3x^2 - 12x + 12$$

$$D = (-12)^2 - 4 \cdot 3 \cdot 12 = 0$$

$$x = \frac{-b}{2a} = \frac{12}{6} = 2$$

$$f'(1) = 3 \Rightarrow y - 0 = 3(x - 1)$$

$$y = 3x - 3$$

$$f''(x) = 6x - 12 \Rightarrow 6x - 12 = 0$$

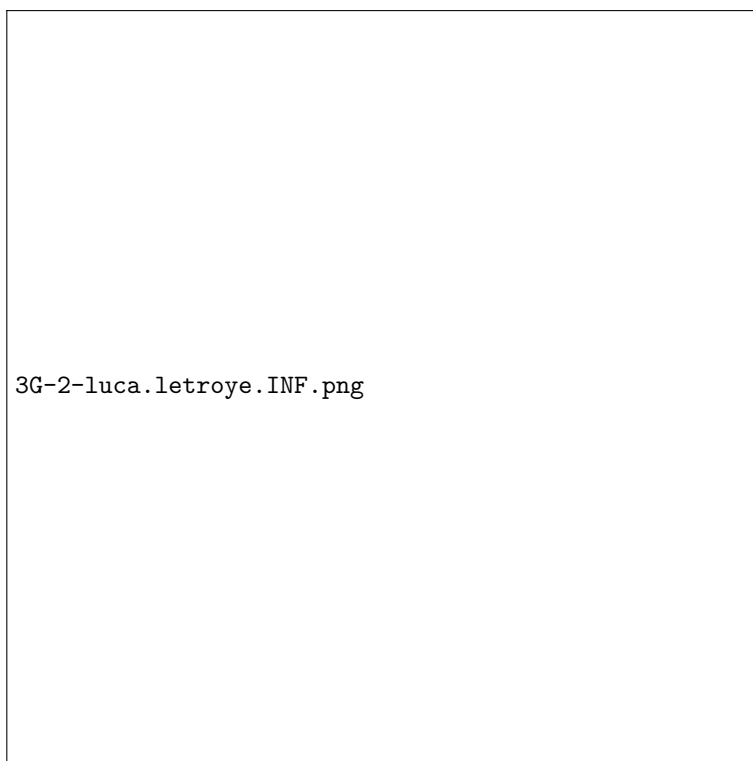
$$x = 2$$

$$\text{dom } f = \mathbb{R}$$

asymptoten: geen

|          | 1          |            |            | 2         |            |
|----------|------------|------------|------------|-----------|------------|
| $f(x)$   | -          | 0          | +          | +         | +          |
| $f'(x)$  | +          | +          | +          | $0^{(2)}$ | +          |
| $f(x)$   | $\nearrow$ | $\nearrow$ | $\nearrow$ | 1         | $\nearrow$ |
| $f''(x)$ | -          | -          | -          | 0         | +          |
| $f(x)$   | $\frown$   | $\frown$   | $\frown$   | B         | $\smile$   |

## References



3G-2-luca.letroye.INF.png